ModelIt!

Teacher's Guide

Why Do Fireworks Explode in Different Colors?

Modeling Electron Energy Transitions and Emission Spectra

Quick Reference

Grade Level	Grade 10L1
NGSS Standard	HS-PS1-1, HS-PS1-2
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-PS1-1, HS-PS1-2

Use the periodic table as a model to predict the relative properties of elements based on the patterns of electrons in the outermost energy level of atoms. Construct and revise an explanation for the outcome of a simple chemical reaction based on the outermost electron states of atoms, trends in the periodic table, and knowledge of the patterns of chemical properties.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop computational models to predict how different metal elements produce specific colors through electron energy transitions during combustion.
Disciplinary Core Idea	**PS1.A Structure and Properties of Matter** — Each atom has a characteristic electron configuration. When electrons transition between energy levels, they emit or absorb photons of specific wavelengths, producing the element's unique emission spectrum.
Crosscutting Concept	**Patterns** — Students identify the pattern that each element's unique electron energy gaps produce a characteristic emission spectrum, enabling identification and prediction of atomic behavior.

CAST Testing Alignment

MULTIPLE CHOICE:

A pyrotechnician adds strontium chloride to a firework shell and observes bright red light at 650nm. If they replace the strontium with copper chloride, what change would the model predict and why?

CONSTRUCTED RESPONSE:

Using your model, explain why every element produces a unique color when heated. Connect your explanation to electron energy levels, photon emission, and the electromagnetic spectrum.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Every Fourth of July, millions of people watch quantum mechanics in action without realizing it. The dazzling colors of fireworks are not produced by dyes or pigments — they are the direct result of electrons inside metal atoms jumping between quantized energy levels and releasing photons of specific wavelengths. Understanding firework colors means understanding the fundamental structure of atoms.

Quantized Electron Energy Levels

In 1913, Niels Bohr proposed that electrons orbit the nucleus only at specific, allowed energy levels — not at arbitrary distances. When an electron absorbs energy, it jumps to a higher level (excitation). When it falls back down, it releases that exact energy difference as a photon of light. The energy gap between levels is unique to each element, which is why each element has a unique emission spectrum.

How Fireworks Produce Color

A firework shell contains fuel (charcoal, sulfur), an oxidizer (potassium perchlorate), and a color-producing metal salt. When the shell detonates, the combustion reaction reaches 1000-2500°C, providing enough thermal energy to excite the metal's electrons. As billions of electrons simultaneously fall back to lower energy levels, they emit photons at the element's characteristic wavelength — strontium emits at ~650nm (red), barium at ~525nm (green), copper at ~450nm (blue).

The Electromagnetic Spectrum and Color

Visible light is a narrow band of the electromagnetic spectrum, from about 380nm (violet) to 700nm (red). The energy of a photon is inversely proportional to its wavelength: $E = hc/\lambda$. This means blue photons carry more energy than red photons. When we see a red firework, we're seeing lower-energy photons from smaller electron energy gaps; blue fireworks release higher-energy photons from larger gaps.

Temperature and Color Purity

Combustion temperature is a critical variable. Too low, and insufficient electrons are excited, producing a dim display. Too high, and the intense heat generates blackbody radiation — a continuous spectrum of white light that washes out the element's characteristic color. Master pyrotechnicians carefully balance oxidizer strength to reach the 'sweet spot' where electron excitation is maximized without excessive blackbody contamination.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify element type, electron energy level, oxidizer strength, combustion temperature, and color wavelength as the key components of the firework color system.

E **Establish Relationships**
Students determine that element type sets the electron energy gap, oxidizer strength drives combustion temperature, and the energy gap determines the photon wavelength (color) emitted when excited electrons fall back down.

V **Visualize & Model**
Students build a computational model showing how metal salt selection and combustion conditions interact to produce specific colors through electron energy transitions.

E **Evaluate Outcomes**
Students run strontium red, copper blue, and temperature override scenarios to test predictions about color output and discover the relationship between energy gaps and photon wavelength.

R **Revise & Extend**
Students add atmospheric oxygen, shell burst pattern, or spectral purity to explore how real-world factors affect color quality and display design.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Every color in that fireworks show is quantum mechanics you can see with your naked eye. Red, blue, green — each one is atoms telling you their secrets."*

Watch For

Show a 15-second fireworks video clip. Ask: Where do you think the colors come from? Is it a dye?

Slide 2: Learning Objectives

3 min

Say: *"Today we're going to decode the atomic science behind every firework you've ever seen."*

Watch For

Have students read objectives. Pre-teach 'emission spectrum' and 'photon' with quick visual aids.

Slide 3: Big Question

3 min

Say: *"The gunpowder is the same. The fuse is the same. The shell is the same. So WHY are the colors different?"*

Watch For

Quick-write: Students hypothesize what makes red different from blue in a firework. Share out.

Slide 4: LEVER Framework

2 min

Say: *"We're going to build a model that predicts what color a firework produces based on what's inside it."*

Watch For

Preview each LEVER step. Emphasize that this is about atomic structure, not just chemistry.

Slide 5: Activity 1: Components

8 min

Say: *"What factors determine the color of a firework? Which ones can a pyrotechnician control?"*

Watch For

Guide sorting of external vs. internal components. Discuss why element type and oxidizer are choices (external).

Slide 6: Activity 2: Connections

8 min

Say: *"When strontium is heated, what happens inside each atom? Follow the energy from explosion to your eye."*

Watch For

Trace the pathway: heat → electron excitation → photon emission → color. Use Bohr model diagrams.

Slide 7: Activity 3: Simulation

10 min

Say: *"Let's test three metals and see if the model can predict the color before we see it."*

Watch For

Students predict colors for strontium, copper, and barium. Run simulations. Then test the temperature override.

Slide 8: Discoveries

5 min

Say: *"So the color isn't in the fire — it's in the atoms. And astronomers use this exact same science to identify what stars are made of."*

Watch For

Show real emission spectra for hydrogen, sodium, and neon. Connect fireworks to stellar spectroscopy.

Slide 9: STEM Challenge

12 min

Say: *"A city just hired your team to design a three-color fireworks show. Pick your metals, set your oxidizers, and prove your colors will work."*

Watch For

Groups select metal salts, determine oxidizer ratios from model data, and present spectral predictions.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Element Type, Oxidizer Strength
- **INTERNAL:** Electron Energy Level, Combustion Temperature, Color Wavelength

Relationships

- **Element Type to Electron Energy Level:** DETERMINES — Each element has a unique set of quantized electron energy levels. The specific metal salt selected defines the energy gaps available for electron transitions, which ultimately determines the color of light emitted.
- **Oxidizer Strength to Combustion Temperature:** POSITIVE (+) — Stronger oxidizers drive more vigorous combustion reactions, producing higher temperatures. Higher temperatures provide more thermal energy to excite electrons to upper energy levels.
- **Electron Energy Level to Color Wavelength:** DETERMINES — The energy gap between the excited level and the ground state determines the exact wavelength of the emitted photon. Larger gaps produce shorter wavelengths (blue), smaller gaps produce longer wavelengths (red).

Key Discoveries

- Each element produces a unique color because its electrons have specific, quantized energy gaps — the color is determined by atomic structure, not the explosion itself
- The color of light corresponds directly to the energy released when an excited electron falls back to a lower level — red light carries less energy per photon than blue light
- Oxidizer strength must be carefully balanced — too little and electrons aren't excited enough, too much and the extreme temperature produces white blackbody radiation that drowns out the element's color
- Firework chemistry is applied quantum mechanics — the same electron transitions that produce firework colors are used by astronomers to identify elements in distant stars

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Firework colors come from dyes or pigments"	There are no dyes in fireworks. The colors are produced by electron transitions within metal atoms. When heated, electrons jump to higher energy levels and then release photons of specific wavelengths as they fall back down. Each element has unique energy gaps, producing unique colors. This is fundamentally different from pigments, which absorb certain wavelengths and reflect others.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Hotter fire always means brighter colors"	Increasing temperature does excite more electrons, but beyond an optimal point, the extreme heat produces blackbody radiation — a continuous white glow that drowns out the element's characteristic color. This is why the hottest stars appear white, not colored. Master pyrotechnicians carefully control temperature to maximize color purity.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"All light is the same — just different shades"	Light is electromagnetic radiation, and different colors correspond to different wavelengths and energies. Red light (~650nm) has longer wavelength and less energy per photon than blue light (~450nm). These are fundamentally different photons carrying different amounts of energy, produced by different-sized electron energy gaps. Color isn't just appearance — it's physics.